

Project-Q Workbook -1

30-day Quantum Computing Challenge

DAY - 01

Introduction to Quantum Computing

Why Should You Care About Quantum Computing?

Imagine trying to solve a puzzle with billions of possible combinations.

A classical computer checks possibilities one by one. As the puzzle grows larger, the time required increases dramatically.

Quantum computers approach certain problems differently by using the laws of quantum mechanics, allowing them to represent and process information in ways that classical systems cannot.

This doesn't mean quantum computers are faster at everything.

Instead:

- Classical Computers = Everyday Computing
- Quantum Computers = Specialized Problem Solvers

Real-World Problems Quantum Computing May Help Solve

Industry	Problem
Healthcare	Drug Discovery
Chemistry	Molecular Simulation
Finance	Portfolio Optimization
Logistics	Route Optimization
Artificial Intelligence	Complex Model Training
Cybersecurity	Advanced Cryptography Research

What is Quantum Computing?

A classical computer stores information using **bits**.

A bit can only be:

0 or 1

A quantum computer stores information using **qubits**.

A qubit can be:

0 1 or a combination of both

This property is called **Superposition**.

Classical Bit vs Quantum Qubit

Classical Bit

CLASSICAL BITS

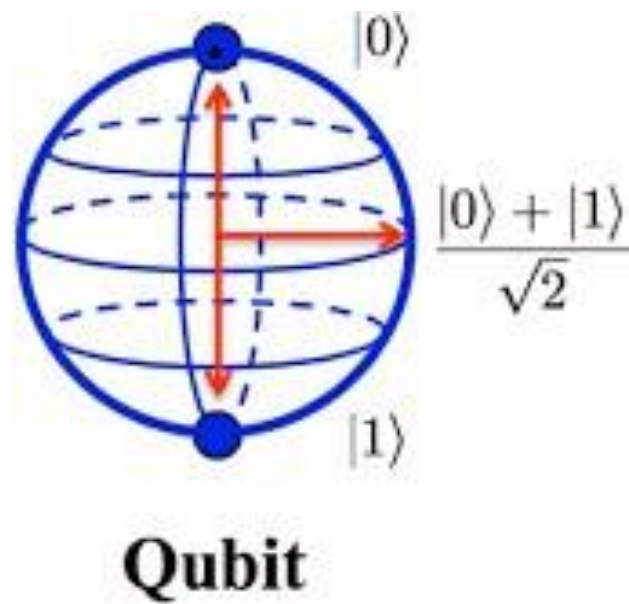


- Definite state
Stores one value at a time



- Foundation of classical computing

Quantum Qubit



A qubit exists as a probability mixture of both states until measurement.

Why Quantum Computing Exists

The Problem

As real-world problems grow larger:

More Variables



More Possibilities



More Computation



Classical Systems Struggle

Example: Drug Discovery

Suppose scientists want to discover a new medicine.

To understand how a drug molecule behaves, they must simulate millions of interactions between atoms and electrons.

A classical computer struggles because:

More Atoms



More Interactions



More Calculations



Exponential Growth

Quantum computers are attractive because nature itself operates using quantum mechanics.

Core Principles of Quantum Computing

1. Qubits

Definition

The fundamental unit of quantum information.

Think of it Like

A spinning coin.

While spinning:

Heads?

Tails?

Both are possible.

Once it lands:

Heads OR Tails

Similarly:

Measurement



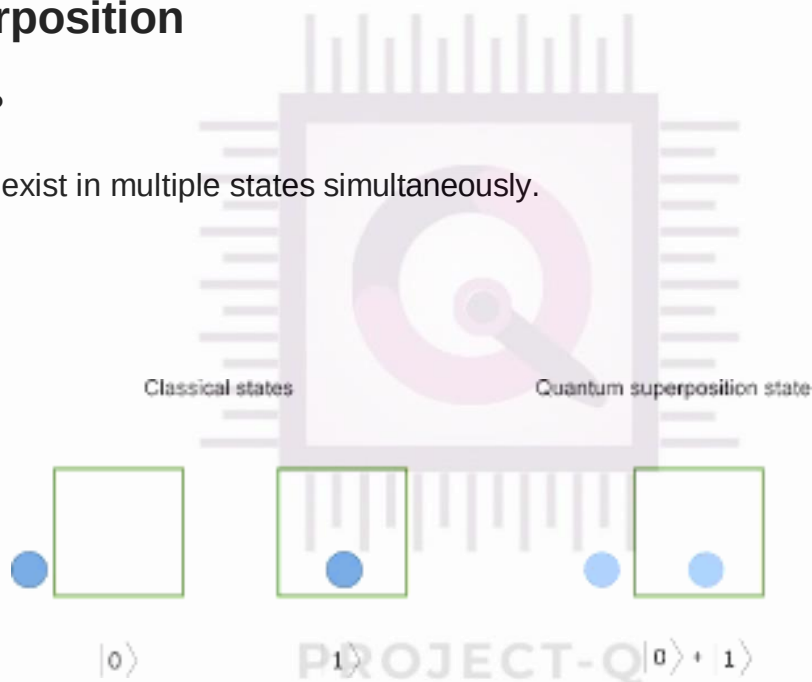
Qubit becomes
0 or 1

2. Superposition

What Is It?

A qubit can exist in multiple states simultaneously.

Visual



The dot can exist
between both states.

Why It Matters

With multiple qubits, the number of possible states grows rapidly.

Example:

Qubit s	Possible States
1	2
2	4

3	8
10	1024

3. Entanglement

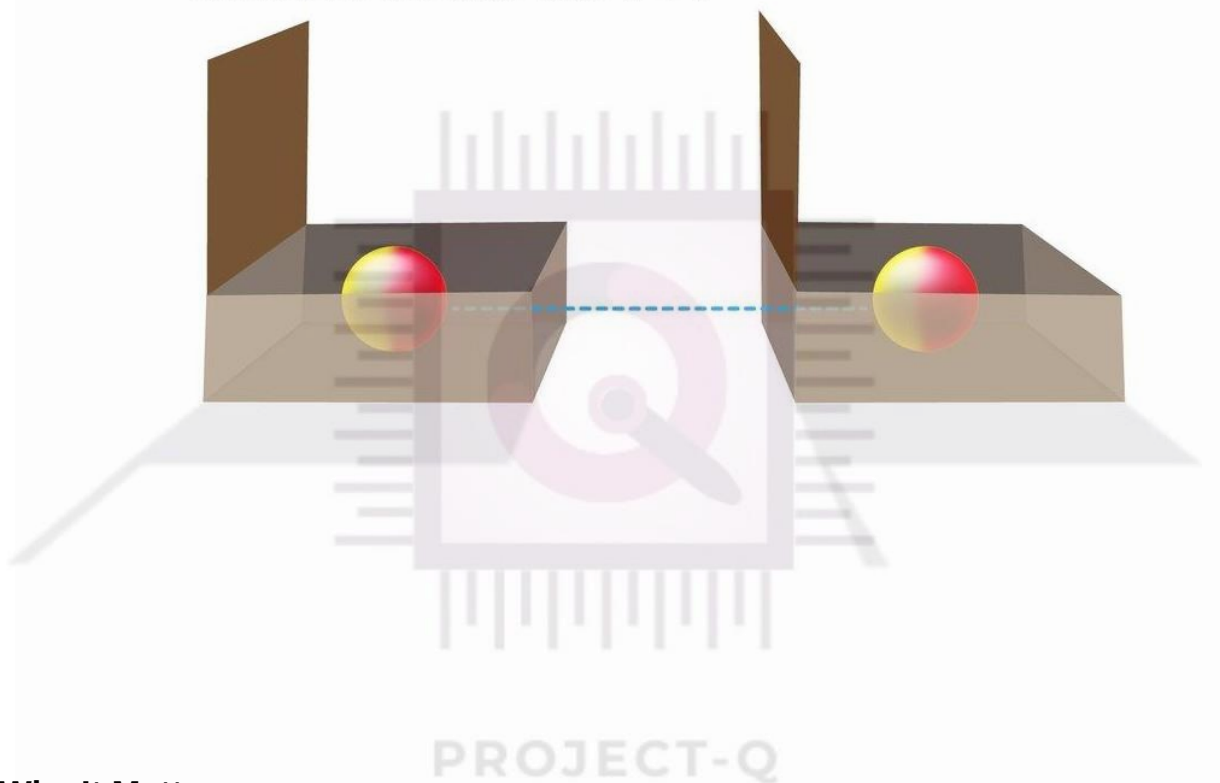
What Is It?

Two qubits become connected.

Changing or measuring one affects information about the other.

Visual

Let's say you have two entangled balls, each in its own box. In this metaphor, the balls can be either yellow or red once observed. For now, they are in a state of superposition, or both yellow and red at the same time . . .



Why It Matters

Enables powerful quantum algorithms.

4. Interference

What Is It?

Quantum algorithms amplify correct answers and suppress incorrect ones.

Visual

Possible Results



Algorithm amplifies C

5. Decoherence

The Biggest Challenge

Quantum information is fragile.

Noise can destroy computations.

Visual

Ideal Quantum State



Environmental Noise



Information Loss



Decoherence

Classical vs Quantum Computing

Feature	Classical	Quantum
Unit	Bit	Qubit
States	0 or 1	0 and 1
Processing	Deterministic	Probabilistic
Noise Sensitivity	Low	High
Everyday Tasks	Excellent	Poor
Optimization Problems	Limited	Potentially Strong

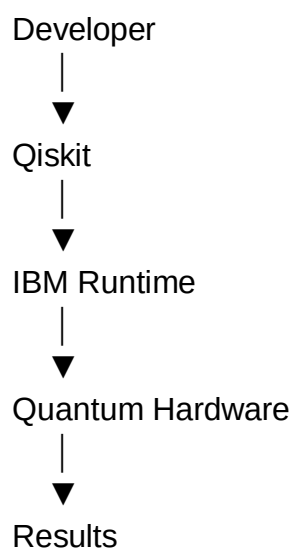
IBM Quantum Learning Ecosystem

Why IBM Matters

IBM provides:

- Real Quantum Hardware
- Cloud Access
- Learning Resources
- Quantum Development Tools

IBM Quantum Stack



What is Qiskit?

Qiskit is IBM's open-source quantum software development kit.

Think of it as:

Python for Quantum Computing

Using Qiskit you can:

- Build Circuits
- Simulate Programs
- Access Real Quantum Computers
- Analyze Results

YouTube References

- Video 1: <https://youtu.be/lt4OsgmUTGI?si=gRXpR6HoYZ1JXyLM>
 - Video 2: <https://youtu.be/B3U1NDUiwSA?si=5Ub1VKglt2sPUgxo>
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PROJECT-Q

End-of-Day Summary

What You Learned Today

- What Quantum Computing Is
- Why Quantum Computing Exists
- Qubits
- Superposition
- Entanglement
- Interference
- Decoherence
- IBM Quantum Learning
- Qiskit